

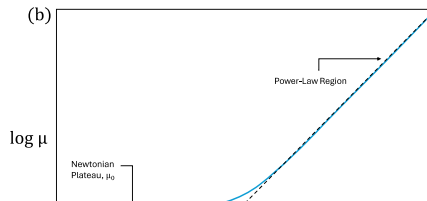
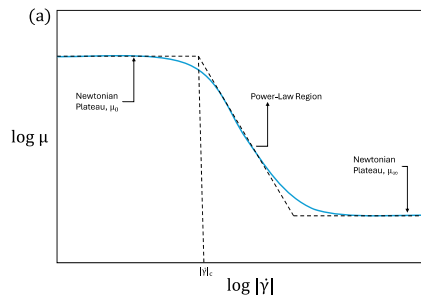
On Self-Similar Viscosity Models for Carreau Boundary Layer Flows

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One-slide version

- Boundary-layer flows have high shear near the wall/leading edge and low shear far from the wall/downstream.
- Power-law and Newtonian models admit similarity solutions; Carreau rheology does not.
- Main question: when do simple self-similar models accurately predict Carreau wall shear?
- Main result: use power-law upstream, Newtonian downstream, with a computable transition criterion and quantified high-shear error.



Roadmap for a 20-minute talk

- ① Boundary-layer setup and viscosity models.
- ② Similarity reductions: power-law/Newtonian versus Carreau.
- ③ New Sakiadis power-law solutions outside the previously accepted range.
- ④ Wall-shear comparison and the transition parameter δ^* .
- ⑤ Why the high-shear power-law approximation has a finite offset.

Why the power-law model is both useful and incomplete

Power-law rheology

$$\mu = K|\dot{\gamma}|^{\alpha-1}$$

- $\alpha < 1$: shear-thinning.
- $\alpha > 1$: shear-thickening.
- Analytically convenient; often self-similar.

Problem

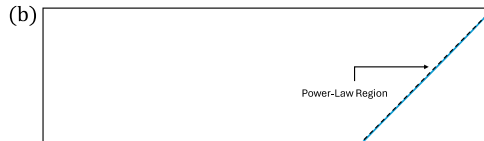
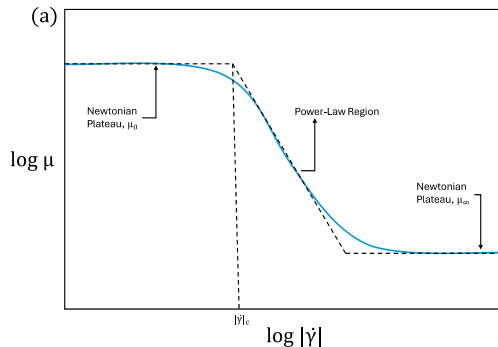
- It has no low-shear Newtonian plateau.
- In a boundary layer, low-shear and high-shear regions coexist in the same flow.
- So the far-field velocity profile can be wrong even when wall shear looks close.

Carreau rheology: power-law behavior with Newtonian regularization

$$\mu = \mu_0 \left(1 + (\lambda|\dot{\gamma}|)^2\right)^{(\alpha-1)/2}$$

$$\mu \sim \mu_0 \quad (|\dot{\gamma}| \rightarrow 0), \quad \mu \sim \mu_0 \lambda^{\alpha-1} |\dot{\gamma}|^{\alpha-1} \quad (|\dot{\gamma}|$$

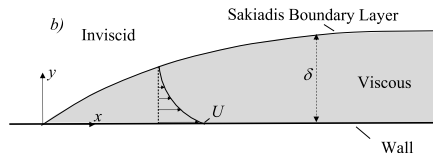
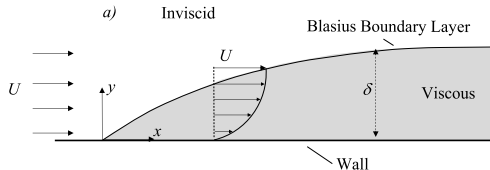
$$|\dot{\gamma}|_c = \frac{1}{\lambda}, \quad K = \mu_0 \lambda^{\alpha-1}.$$



Two canonical boundary layers

- **Blasius:** stationary wall, moving fluid.
- **Sakiadis:** moving wall, quiescent fluid.
- They are not equivalent by a frame shift because the governing boundary-layer equation is nonlinear.

Location	Blasius	Sakiadis
$y = 0$	$u_x = 0$	$u_x = U$
$y \rightarrow \infty$	$u_x \rightarrow U$	$u_x \rightarrow 0$



Boundary-layer equations and comparison metric

$$\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} = 0,$$

$$\rho \left(u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} \right) = \frac{\partial \tau_{yx}}{\partial y},$$

$$\tau_{yx} = \mu(|\dot{\gamma}|) \frac{\partial u_x}{\partial y}, \quad |\dot{\gamma}| = \left| \frac{\partial u_x}{\partial y} \right|.$$

Metric

Wall shear / wall shear-rate is the central quantity:

$$\tau_i \quad (i = n, p, c).$$

It is sensitive to the full boundary-value problem, but is still the most engineering-facing scalar output.

Power-law and Newtonian models are self-similar

$$\eta = y \left(\frac{\rho U^{2-\alpha}}{K_X} \right)^{1/(\alpha+1)}, \quad u_x = U f'_p(\eta; \alpha).$$

$$\alpha(\alpha + 1) f_p''' + f_p f_p'' |f_p''|^{1-\alpha} = 0.$$

Boundary conditions

Blasius:

$$f_p(0) = 0, \quad f_p'(0) = 0, \quad f_p'(\infty) = 1.$$

Sakiadis:

$$f_p(0) = 0, \quad f_p'(0) = 1, \quad f_p'(\infty) = 0.$$

Carreau is not self-similar, but reduces to a one-parameter family

Use the Newtonian similarity scaling:

$$f_c = \psi \sqrt{\frac{\rho}{\mu_0 x U}}, \quad \eta = y \sqrt{\frac{\rho U}{\mu_0 x}}.$$

The remaining streamwise dependence enters through

$$\delta = \frac{\lambda^2 U^3 \rho}{\mu_0 x} = \frac{\text{De}^2}{\text{Re}_{n,x}}.$$

$$(1 + \alpha \delta f_c''^2) (1 + \delta f_c''^2)^{(\alpha-3)/2} f_c''' + \frac{1}{2} f_c f_c'' = 0$$

- Solve an ODE at each value of δ .
- Large x means small δ : Newtonian behavior.
- Near the leading edge, δ is large: power-law behavior.

New Sakiadis power-law solutions: what was missing

- Earlier work concluded that Sakiadis power-law solutions fail for $\alpha < 1/2$ and $\alpha > 1$.
- That conclusion depends on assuming a far-field form with $f_p \rightarrow C$.
- The boundary condition only requires $f'_p \rightarrow 0$; it does not require f_p to approach a constant.
- Alternative far-field structures produce admissible separatrix solutions.

These new solutions are essential for comparing Sakiadis power-law predictions to Carreau predictions across shear-thinning and shear-thickening regimes.

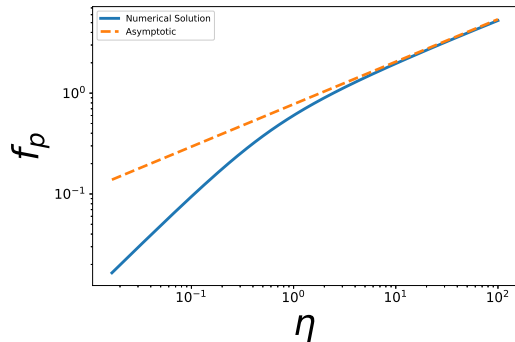
Shear-thinning Sakiadis solutions for $0 < \alpha < 1/2$

The new far-field balance is

$$f_p(\eta; \alpha) \sim H(\eta + J)^p, \quad p = \frac{1 - 2\alpha}{2 - \alpha}.$$

$$H = \left(\frac{\alpha(\alpha + 1)(2 - p)}{p^{1-\alpha}(1 - p)^{1-\alpha}} \right)^{1/(2-\alpha)}.$$

- Then $0 < p < 1$, so $f'_p \rightarrow 0$ while f_p grows sublinearly.
- This satisfies the Sakiadis far-field velocity condition.



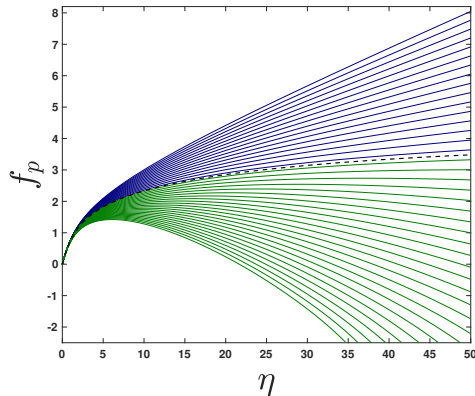
The admissible solution is a separatrix

Shoot with

$$f_p''(0; \alpha) = \kappa.$$

There is a critical value $\kappa = \kappa_p$ such that:

- above it, solutions remain monotone;
- below it, solutions turn over;
- at it, $f_p' \rightarrow 0$ smoothly.



Shear-thickening Sakiadis solutions for $\alpha > 1$

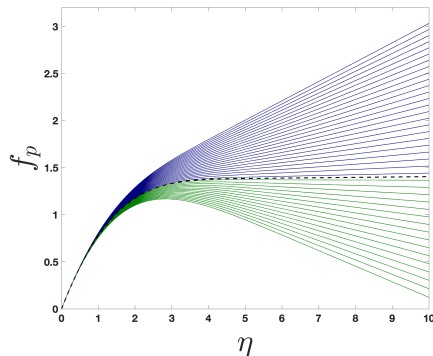
For $\alpha > 1$, the separatrix has finite thickness:

$$f_p(\eta; \alpha) = C_2 \quad \text{for } \eta \geq C_1.$$

Consequently,

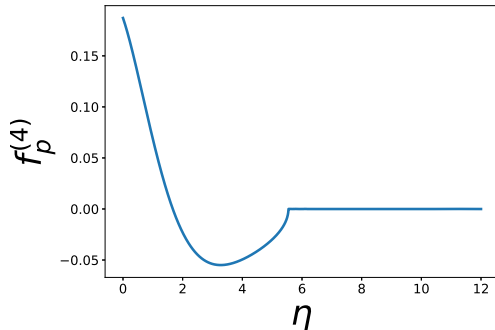
$$u_x = Uf'_p(\eta; \alpha) = 0 \quad \text{outside the boundary layer.}$$

The solution is C^4 but not C^5 at the boundary-layer edge.



Finite-width thickening solutions have a fifth-derivative jump

- Velocity, velocity gradient, and curvature remain continuous.
- The fifth derivative jumps at the edge of the finite-thickness layer.
- This mirrors the known singular structure in shear-thickening Blasius flow.
- The Carreau model smooths this because it recovers Newtonian viscosity at low shear.



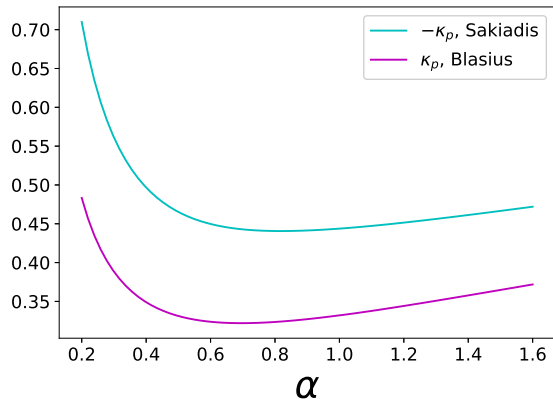
The new branches connect continuously in wall shear

Define the dimensionless wall-shear coefficient

$$\kappa_p = f_p''(0; \alpha).$$

The computed values vary continuously through:

- the known range $1/2 < \alpha \leq 1$,
- the new shear-thinning range $\alpha < 1/2$,
- the new shear-thickening range $\alpha > 1$.



Wall-shear scalings for comparison

With $\bar{\tau}_i = \tau_i/(\rho U^2)$,

$$\bar{\tau}_n = \kappa_n \text{Re}_{n,x}^{-1/2}.$$

$$\bar{\tau}_p = \kappa_p^\alpha \text{De}^{(\alpha-1)/(\alpha+1)} \text{Re}_{n,x}^{-\alpha/(\alpha+1)}.$$

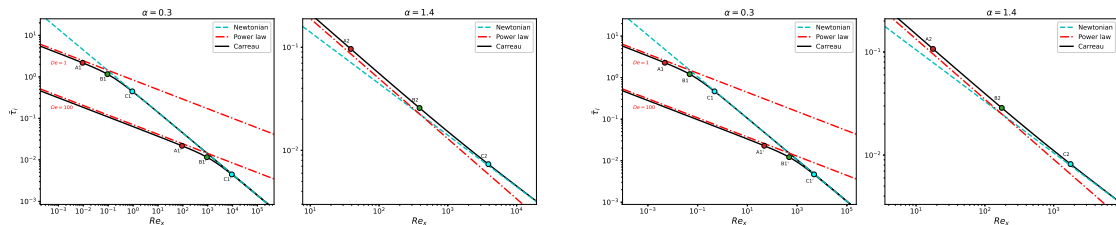
$$\bar{\tau}_c = \kappa_c \text{Re}_{n,x}^{-1/2} (1 + \kappa_c^2 \delta)^{(\alpha-1)/2}, \quad \kappa_c = f_c''(0)$$

$$\delta = \frac{\text{De}^2}{\text{Re}_{n,x}}.$$

Interpretation

Changing x changes $\text{Re}_{n,x}$ and hence changes which part of the rheological curve the boundary layer samples.

Carreau wall shear follows the two self-similar asymptotes



- Near the leading edge: high shear, Carreau follows the power-law similarity prediction.
- Far downstream: low shear, Carreau approaches Newtonian wall shear.
- The transition occurs where the Newtonian and power-law asymptotes cross.

Asymptotic agreement and a finite high-shear correction

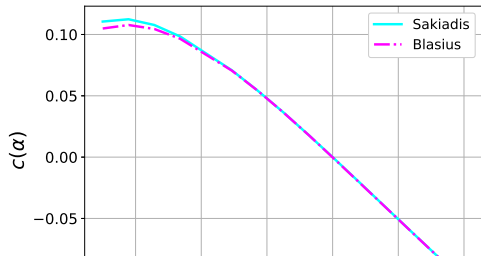
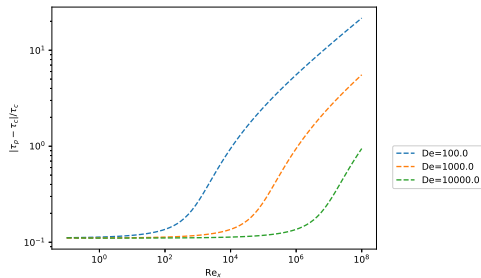
Downstream:

$$\bar{\tau}_c \sim \bar{\tau}_n \quad \text{as } \text{Re}_{n,x} \rightarrow \infty.$$

Near the leading edge:

$$\bar{\tau}_c \sim (1 - c(\alpha))\bar{\tau}_p \quad \text{as } \text{Re}_{n,x} \rightarrow 0.$$

- The correction is independent of De in the high-shear limit.
- For $\alpha \in [0.3, 1.4]$, the high-shear offset is roughly at most 10–11%.



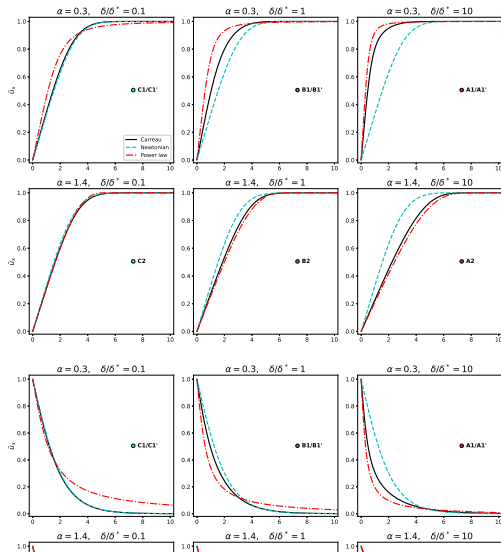
Why is there any high-shear error?

The wall shear is not purely local. For Blasius,

$$\bar{\tau}_i = \frac{d}{dx} \int_0^\infty u_x(U - u_x) dy, \quad i = n, c, p.$$

For Sakiadis, use the analogous expression with $U = 0$ in the far field.

- The full velocity profile contributes.
- Power-law and Carreau profiles can have similar near-wall slopes but different far-field behavior.
- That profile difference produces $c(\alpha)$.



A practical criterion: when to switch models

Set the Newtonian and power-law wall-shear predictions equal. The crossover occurs at

$$\delta^* = \left(\frac{\kappa_p^\alpha}{\kappa_n} \right)^{2(1+\alpha)/(1-\alpha)}.$$

Rule of thumb

- $\delta/\delta^* \gg 1$: use the power-law similarity solution.
- $\delta/\delta^* \ll 1$: use the Newtonian similarity solution.
- Transition region: solve the Carreau one-parameter ODE if accuracy matters.

- δ^* is relatively insensitive to α : roughly 14–20 over a broad range.
- Drag estimates can be obtained by splitting the plate at the x corresponding to $\delta = \delta^*$.

Conclusions

- ① Carreau wall shear is well captured by self-similar Newtonian and power-law asymptotes outside the transition region.
- ② The high-shear power-law approximation has a finite, quantified correction $c(\alpha)$ caused by full-profile differences.
- ③ New Sakiadis power-law similarity solutions exist for $\alpha < 1/2$ and $\alpha > 1$.
- ④ For $\alpha > 1$, the Sakiadis solution is finite-width and C^4 but not C^5 at its edge.
- ⑤ The crossover parameter δ^* gives a low-cost practical approximation framework for non-self-similar Carreau boundary layers.

Backup: minimal numerical recipe

- Convert the relevant BVP to a shooting IVP by prescribing $f''(0)$.
- Truncate $\eta \in [0, \infty)$ to $[0, L]$.
- Use RK45 integration and choose $f''(0)$ so that the far-field condition is met at $\eta = L$.
- Increase L until wall-shear coefficients are converged to the desired tolerance.

This is cheap for power-law/Newtonian models and still manageable for Carreau because only the parameter δ is swept.